

Project Report

Design of a Battery Energy Storage System with Grid Integration

Submitted by

Ashwin Shankar

Department of Electrical Engineering
Manipal Institute of Technology

Date: June 7, 2026

Contents

Abstract	3
1 Introduction	4
2 Objectives	5
3 System Description	6
3.1 Open-Loop System Diagram	6
3.2 PV Array	7
3.3 Boost Converter	7
3.4 Battery Pack	8
3.5 Bidirectional DC-DC Converter	8
3.6 DC-Link Capacitor	9
3.7 Three-Phase Voltage Source Inverter	9
3.8 Grid-Side Filter	10
3.9 Three-Phase Grid	10
3.10 Control System	10
3.10.1 PV Array Data	11
3.11 Battery Pack Data	11
3.12 Power Flow in Different Operating States	12
3.12.1 State 1: Charging	12
3.12.2 State 2: Discharging	12
3.12.3 State 3: Standstill	13
4 Control Strategy	14
4.1 Mode Selection Logic	14
4.2 Need for dq-Axis Control	14
5 Simulation Model	16
6 Results and Discussion	17
6.1 State 1: Battery Charging	17

6.1.1	Grid-Side Outputs	17
6.1.2	Boost Converter Outputs	18
6.1.3	Battery SOC, Current, and Voltage	18
6.2	State 2: Battery Discharging	19
6.2.1	Grid-Side Outputs	19
6.2.2	Boost Converter Outputs	20
6.2.3	Battery SOC, Current, and Voltage	20
6.3	State 3: Battery Standstill Condition	21
6.3.1	Grid-Side Outputs	21
6.3.2	Boost Converter Outputs	22
6.3.3	Battery SOC, Current, and Voltage	22
7	Observations	23
8	Need for dq Control	24
8.1	dq Control Implementation for Grid Synchronization	25
8.1.1	Closed-Loop dq Control Structure	25
8.1.2	PI Current Controller	26
8.1.3	Grid-Side Voltage and Current Observation	27
8.1.4	Reason for Lack of Synchronization	27
8.1.5	Expected Result After dq Control	28
9	Applications	29
10	Advantages and Limitations	30
10.1	Advantages	30
10.2	Limitations	30
11	Conclusion	31

Abstract

This report presents the design and simulation of a Battery Energy Storage System integrated with the grid. The system is designed to operate in three major modes: battery charging, battery discharging, and battery standstill condition. The proposed system consists of a battery pack, bidirectional DC-DC converter, DC-link, three-phase inverter, filter, and grid connection.

The main objective of the system is to control the power flow between the battery and the grid depending on the operating condition. During charging mode, power is transferred from the grid to the battery. During discharging mode, the battery supplies power to the grid or load. During standstill mode, the battery remains isolated or inactive, and no major power exchange takes place.

Simulation results are analyzed for all three operating states. The results include grid-side voltage and current waveforms, boost converter voltage and current outputs, and battery parameters such as State of Charge, battery current, and battery voltage. It was observed that the grid current and inverter current were not synchronized properly. Therefore, dq-axis control is required for accurate grid synchronization, active and reactive power control, and stable grid-connected operation.

Chapter 1

Introduction

Battery Energy Storage Systems are widely used in modern power systems for energy management, renewable energy integration, grid support, and backup power applications. With the increasing penetration of renewable energy sources, energy storage systems play an important role in maintaining power balance and improving system reliability.

A grid-integrated battery energy storage system allows bidirectional power flow between the grid and the battery. When excess power is available, the battery can be charged. When power demand increases or grid support is required, the battery can discharge power into the grid or load.

The system studied in this project operates in three states:

- State 1: Battery charging condition
- State 2: Battery discharging condition
- State 3: Battery standstill condition

The performance of the system is analyzed using simulation waveforms obtained from the grid side, converter side, and battery side.

Chapter 2

Objectives

The main objectives of this project are:

- To design a grid-integrated Battery Energy Storage System.
- To model the battery, bidirectional converter, inverter, and grid interface.
- To analyze the system under charging, discharging, and standstill conditions.
- To observe grid-side voltage and current behavior during different operating states.
- To analyze boost converter output voltage and current.
- To monitor battery State of Charge, battery current, and battery voltage.
- To identify the need for dq-axis control for proper grid synchronization.

Chapter 3

System Description

The proposed system is a grid-integrated Battery Energy Storage System supported by a photovoltaic source. The system consists of a PV array, DC-DC boost converter, battery pack, bidirectional converter, DC-link capacitor, three-phase voltage source inverter, grid-side filter, and three-phase grid.

The purpose of the system is to manage power flow between the PV source, battery pack, and grid. During high irradiance conditions, the PV array supplies power to the DC-link and can also charge the battery. During low irradiance or night-time conditions, the battery supplies power to the inverter. During standstill condition, the battery is neither charged nor discharged.

3.1 Open-Loop System Diagram

The open-loop system configuration is shown in Fig. 3.1. In this system, the switching pulses are generated without closed-loop feedback control. The PV array is connected to the DC-link through a boost converter. The battery pack is connected through a bidirectional DC-DC converter. The DC-link feeds a three-phase inverter, which is connected to the grid through an output filter.

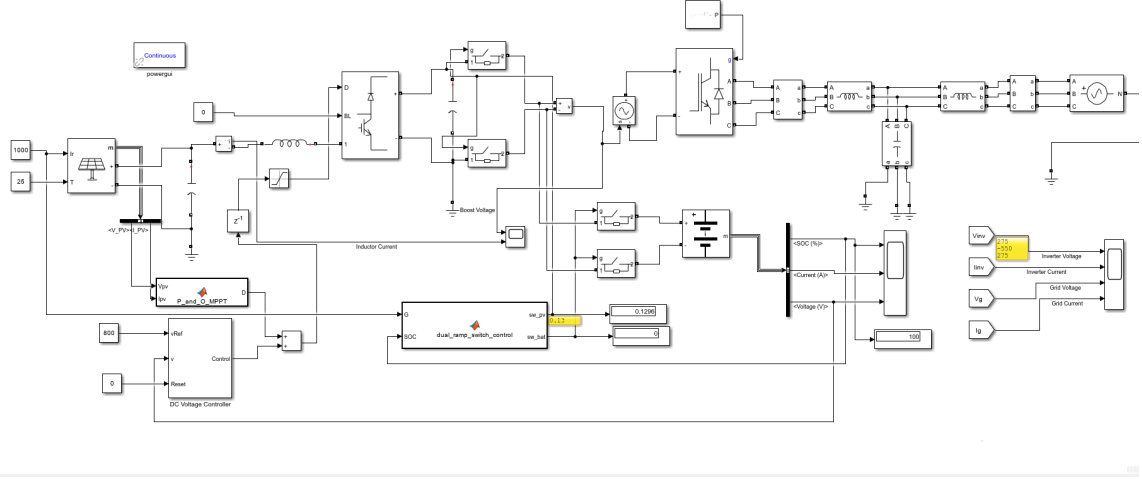


Figure 3.1: Open-loop system diagram of PV-assisted Battery Energy Storage System with grid integration

The open-loop system is mainly used to observe the basic operating behavior of the system under different states. However, open-loop operation does not provide accurate grid synchronization, DC-link regulation, or controlled active and reactive power exchange. Hence, closed-loop dq-axis control is required for improved grid-connected performance.

3.2 PV Array

The photovoltaic array acts as the renewable energy source of the system. It converts solar irradiance into DC electrical power. The output voltage and current of the PV array depend mainly on irradiance and temperature.

During high irradiance conditions, the PV array generates sufficient power and supplies energy to the DC-link through the boost converter. During low irradiance conditions, the PV output power reduces and the battery may be used to support the DC-link.

The output power of the PV array is given by:

$$P_{PV} = V_{PV} I_{PV}$$

where V_{PV} is the PV voltage and I_{PV} is the PV current.

3.3 Boost Converter

The boost converter is used to step up the PV voltage to the required DC-link voltage level. Since the PV array voltage may be lower than the required DC-link voltage, a boost converter is necessary.

For an ideal boost converter, the output voltage is given by:

$$V_{dc} = \frac{V_{PV}}{1 - D}$$

where D is the duty cycle of the boost converter.

In this project, the boost converter output is connected to the DC-link. The duty cycle determines the amount of voltage boosting. Under open-loop operation, the duty cycle is fixed or externally controlled. In closed-loop operation, the duty cycle can be adjusted using MPPT or voltage control.

3.4 Battery Pack

The battery pack is used as the energy storage element. It stores excess energy during charging mode and supplies energy during discharging mode.

The battery operates in three main conditions:

- Charging condition: Battery receives power and SOC increases.
- Discharging condition: Battery supplies power and SOC decreases.
- Standstill condition: Battery current is approximately zero and SOC remains constant.

The main battery quantities observed in the simulation are:

- Battery State of Charge
- Battery voltage
- Battery current

The State of Charge indicates the available charge in the battery. If the battery current is positive, it may indicate charging depending on the sign convention used in the simulation. If the battery current is negative, it may indicate discharging. The exact sign depends on the battery block convention.

3.5 Bidirectional DC-DC Converter

The bidirectional DC-DC converter is placed between the battery pack and the DC-link. It allows power flow in both directions.

During charging mode, power flows from the DC-link to the battery. In this case, the converter operates in buck mode because the DC-link voltage is usually higher than the battery voltage.

During discharging mode, power flows from the battery to the DC-link. In this case, the converter operates in boost mode because the battery voltage must be raised to the DC-link voltage level.

Therefore, the bidirectional converter performs two functions:

- Controls battery charging current.
- Regulates DC-link support during battery discharging.

3.6 DC-Link Capacitor

The DC-link capacitor is connected between the DC-DC converter and the inverter. It acts as an intermediate energy storage element.

The main functions of the DC-link capacitor are:

- Maintain a stable DC-link voltage.
- Reduce DC voltage ripple.
- Provide temporary energy during switching transients.
- Decouple the DC-DC converter side and inverter side.

A stable DC-link voltage is important because the inverter requires a nearly constant DC input to generate proper AC output voltage.

3.7 Three-Phase Voltage Source Inverter

The three-phase voltage source inverter converts DC power from the DC-link into three-phase AC power. This AC power is then supplied to the grid through the filter.

During battery discharging mode, the inverter injects power into the grid. During battery charging mode, the inverter may operate as an active rectifier, allowing power to flow from the grid to the DC-link.

The inverter output voltage depends on the DC-link voltage and the modulation index. In open-loop operation, the inverter switching pulses are generated without feedback synchronization. Because of this, the inverter current may not be synchronized with the grid current.

3.8 Grid-Side Filter

The grid-side filter is used between the inverter and the grid. The inverter output contains switching harmonics due to PWM operation. The filter reduces these harmonics and improves the quality of current supplied to the grid.

Common filters used in grid-connected inverter systems are:

- L filter
- LC filter
- LCL filter

An LCL filter provides better harmonic attenuation compared to a simple L filter, but it requires proper damping and control design.

3.9 Three-Phase Grid

The three-phase grid acts as the AC power network to which the inverter is connected. The grid can either receive power from the battery system or supply power to charge the battery.

For proper grid-connected operation, the inverter voltage and current must be synchronized with the grid voltage. This requires a Phase Locked Loop and dq-axis current control.

In this project, it was observed that the grid current and inverter current were not synchronized. Therefore, open-loop control is not sufficient for grid-connected operation.

3.10 Control System

The control system decides the operating mode of the system. It determines whether the battery should charge, discharge, or remain in standstill condition.

The basic operating logic is:

- During high irradiance, PV source supplies the DC-link.
- During low irradiance and sufficient battery SOC, battery supplies the DC-link.
- When battery SOC is high, charging is stopped.
- During standstill condition, battery current is maintained close to zero.

In open-loop operation, the switches and duty cycles are controlled using fixed or simple logic-based signals. However, for grid-connected operation, dq control is required to synchronize the inverter current with the grid voltage.

3.10.1 PV Array Data

The photovoltaic array used in the simulation consists of 40 parallel strings with 8 series-connected modules per string. The selected PV module is the Ablytek 6PN6A225-A0 module. The PV array parameters used in the simulation are shown in Table 3.1.

Table 3.1: PV Array Parameters

Parameter	Value
PV module	Ablytek 6PN6A225-A0
Parallel strings	40
Series-connected modules per string	8
Maximum power per module	224.956 W
Cells per module	60
Open-circuit voltage, V_{oc}	36.24 V
Short-circuit current, I_{sc}	8.04 A
Voltage at maximum power point, V_{mp}	30.24 V
Current at maximum power point, I_{mp}	7.44 A
Temperature coefficient of V_{oc}	$-0.3624\ \%/^{\circ}\text{C}$
Temperature coefficient of I_{sc}	$0.054801\ \%/^{\circ}\text{C}$
Irradiance values	1000, 500, 100 W/m ²
Operating temperature	25 $^{\circ}\text{C}$

3.11 Battery Pack Data

The battery pack is used to store energy and support the DC-link when PV power is insufficient. The battery data used in the simulation is shown in Table 3.2.

Table 3.2: Battery pack data

Parameter	Value
Battery nominal voltage	690 V
Initial State of Charge	80%
Battery operating mode 1	Charging
Battery operating mode 2	Discharging
Battery operating mode 3	Standstill
Charging condition	Battery current flows into battery
Discharging condition	Battery current flows out of battery
Standstill condition	Battery current approximately zero

During charging mode, the battery SOC increases. During discharging mode, the battery SOC decreases. During standstill condition, the SOC remains almost constant because there is no significant current flowing into or out of the battery.

The battery power can be expressed as:

$$P_{bat} = V_{bat}I_{bat}$$

where V_{bat} is the battery voltage and I_{bat} is the battery current.

The sign of I_{bat} determines whether the battery is charging or discharging. Depending on the simulation block convention, positive current may represent charging or discharging. Therefore, the sign convention must be verified from the battery model used in Simulink.

3.12 Power Flow in Different Operating States

3.12.1 State 1: Charging

In charging mode, power flows from the grid or PV source to the battery.

$$Grid/PV \rightarrow DC-link \rightarrow Battery$$

In this state:

- Battery current indicates charging.
- Battery SOC increases.
- Battery voltage gradually changes based on charging behavior.
- Converter controls power flow into the battery.

3.12.2 State 2: Discharging

In discharging mode, power flows from the battery to the DC-link and then to the grid.

$$Battery \rightarrow DC-link \rightarrow Inverter \rightarrow Grid$$

In this state:

- Battery current indicates discharging.
- Battery SOC decreases.
- DC-link is supported by the battery.
- Inverter supplies power to the grid.

3.12.3 State 3: Standstill

In standstill mode, the battery is neither charging nor discharging.

$$P_{battery} \approx 0$$

In this state:

- Battery current is approximately zero.
- Battery SOC remains constant.
- Battery voltage remains nearly steady.
- No significant power exchange occurs.

Chapter 4

Control Strategy

The control system is responsible for managing power flow between the battery and the grid. The converter control determines whether the battery should charge, discharge, or remain in standstill condition.

4.1 Mode Selection Logic

The system operates based on the selected operating state:

Table 4.1: Operating state selection

State	Mode	Power Flow	Battery Condition
State 1	Charging	Grid to Battery	SOC increases
State 2	Discharging	Battery to Grid	SOC decreases
State 3	Standstill	No major power flow	SOC constant

4.2 Need for dq-Axis Control

During simulation, it was observed that the grid current and inverter current were not properly synchronized. This means that the inverter output current was not accurately aligned with the grid voltage.

For grid-connected inverter operation, synchronization is very important. Without proper synchronization, the system may produce distorted current, poor power factor, and unstable power transfer.

Therefore, dq-axis control is required.

The dq control method converts three-phase AC quantities into rotating reference frame quantities. This makes it easier to control active and reactive power independently.

In dq control:

$$i_d$$

controls active power, and

$$i_q$$

controls reactive power.

For unity power factor operation:

$$i_q^* = 0$$

The active power reference is controlled using:

$$i_d^*$$

Therefore, dq control allows:

- Grid synchronization
- Active power control
- Reactive power control
- Unity power factor operation
- Improved inverter current quality
- Stable grid-connected operation

Chapter 5

Simulation Model

The simulation model consists of a battery, bidirectional DC-DC converter, DC-link capacitor, inverter, filter, and three-phase grid. The model was tested under three operating states: charging, discharging, and standstill.

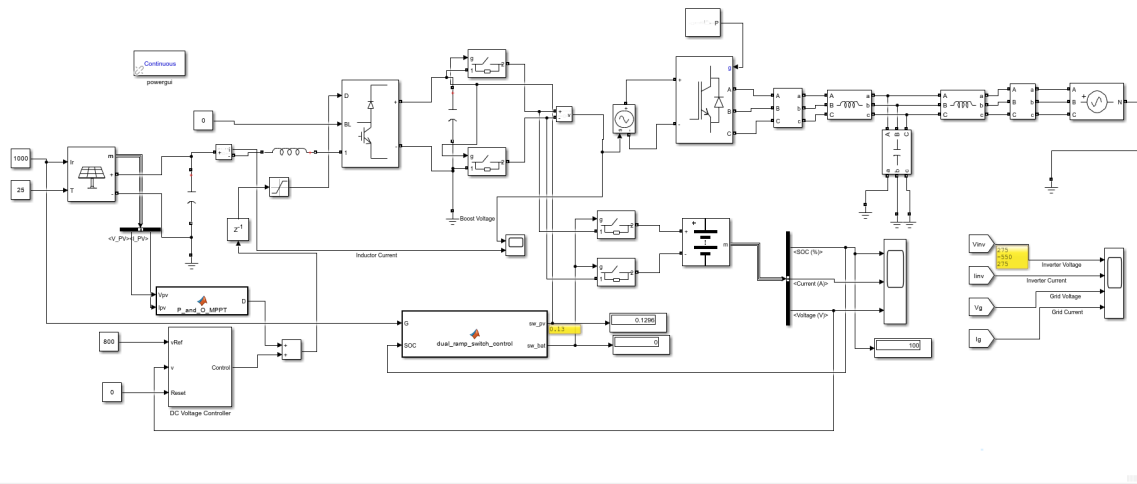


Figure 5.1: Simulation model of Battery Energy Storage System with grid integration

Chapter 6

Results and Discussion

The simulation results are divided into three operating states.

6.1 State 1: Battery Charging

In the charging state, power is transferred from the grid to the battery. The grid supplies power through the inverter and DC-DC converter. The battery current indicates charging operation and the SOC increases gradually.

6.1.1 Grid-Side Outputs

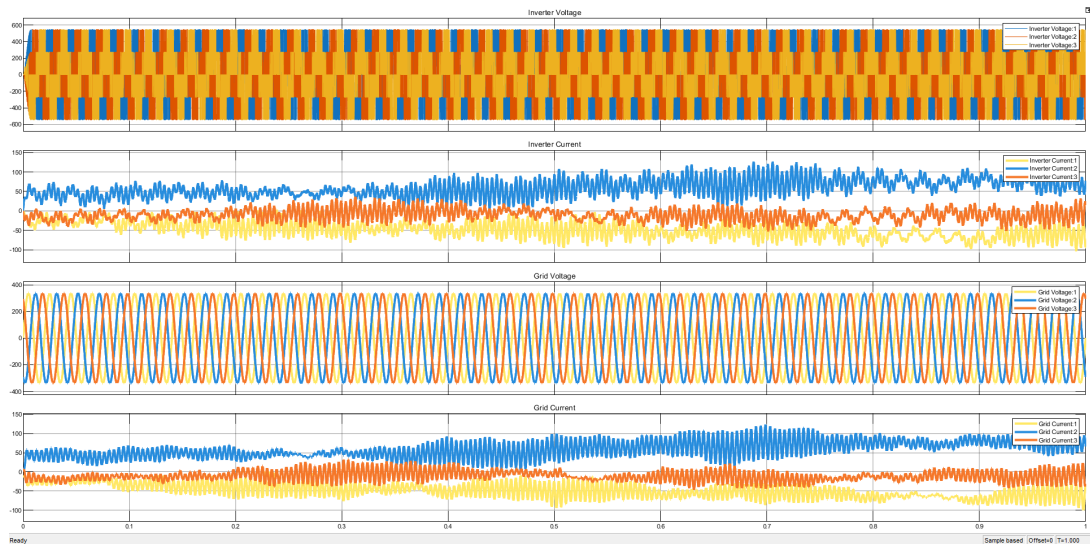


Figure 6.1: State 1: Grid-side voltage and current outputs during battery charging

During charging mode, the grid-side voltage remains sinusoidal. The current drawn from the grid depends on the charging power requirement. Ideally, the grid current should be synchronized with the grid voltage for unity power factor operation.

However, it was observed that the grid current and inverter current were not synchronized properly. This indicates the need for dq-axis current control.

6.1.2 Boost Converter Outputs

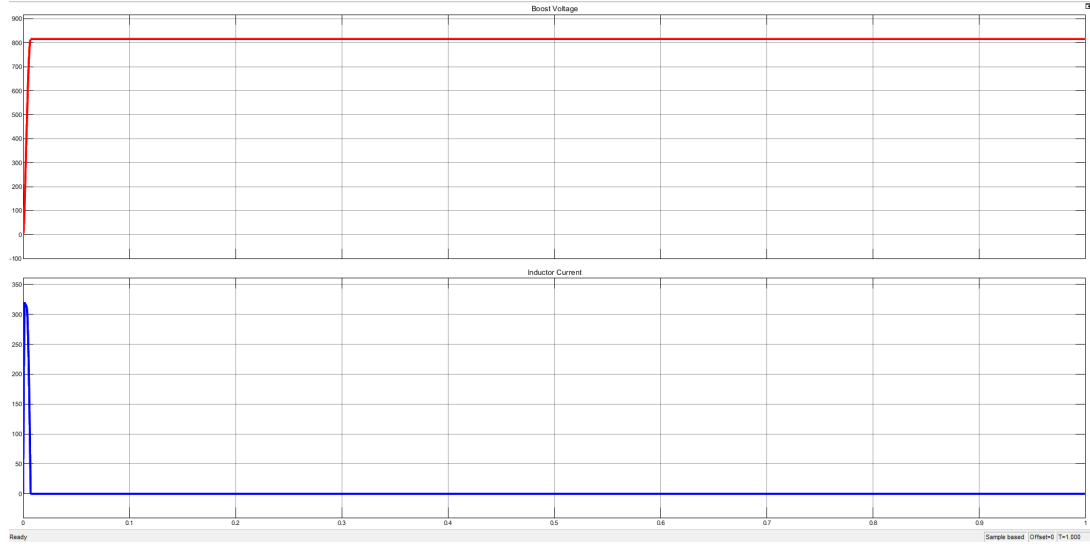


Figure 6.2: State 1: Boost converter voltage and current outputs during battery charging

The converter regulates the power transfer between the DC-link and the battery. During charging, the converter operates in a mode suitable for transferring power from the DC-link to the battery.

6.1.3 Battery SOC, Current, and Voltage

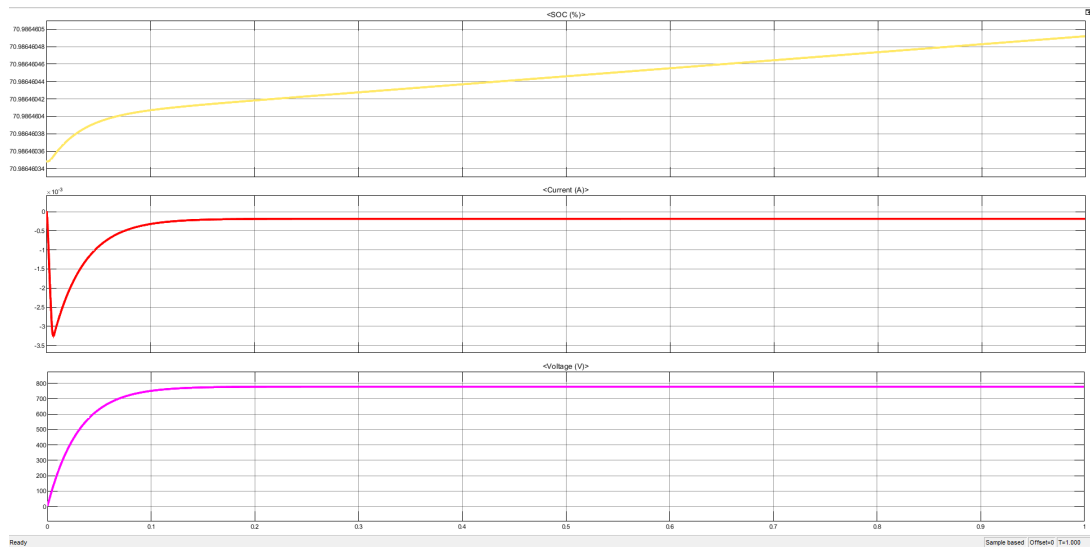


Figure 6.3: State 1: Battery SOC, current, and voltage during charging

During charging, the battery SOC increases with time. The battery current direction corresponds to charging operation. The battery voltage also changes depending on the battery charging characteristics.

6.2 State 2: Battery Discharging

In the discharging state, power is transferred from the battery to the grid. The battery supplies power through the DC-DC converter to the DC-link. The inverter converts DC power into AC power and delivers it to the grid.

6.2.1 Grid-Side Outputs

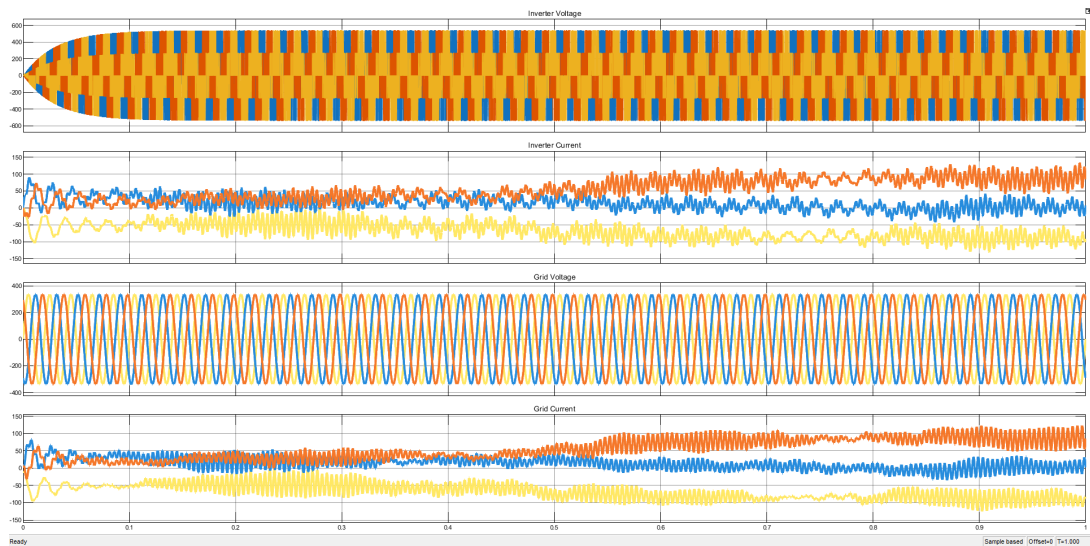


Figure 6.4: State 2: Grid-side voltage and current outputs during battery discharging

During discharging mode, the inverter injects current into the grid. The grid current should be synchronized with the grid voltage for proper power injection. The observed mismatch between grid current and inverter current shows that open-loop or basic control is not sufficient for grid-connected operation.

6.2.2 Boost Converter Outputs

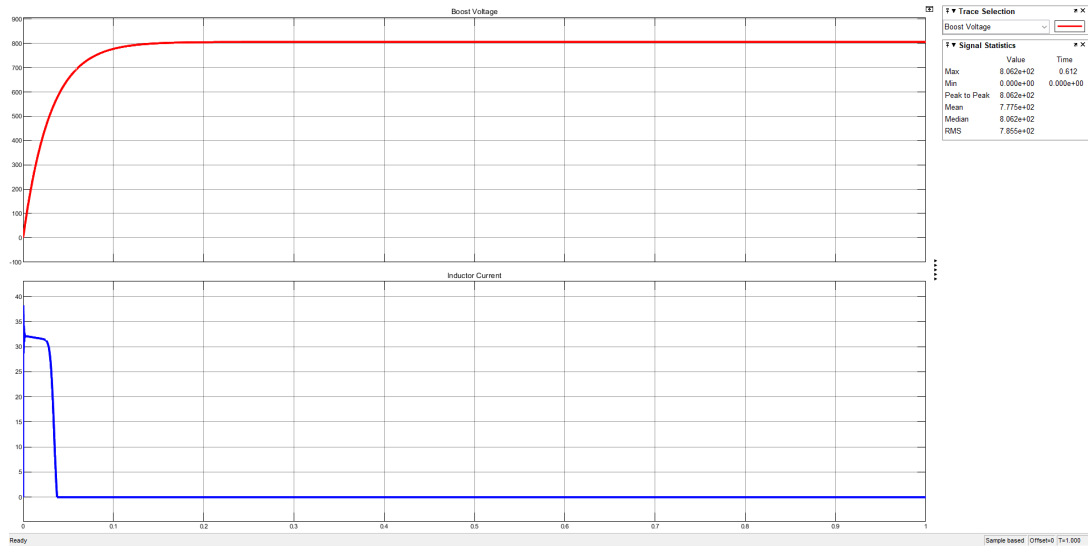


Figure 6.5: State 2: Boost converter voltage and current outputs during battery discharging

During discharging mode, the converter regulates the DC-link voltage by transferring energy from the battery to the inverter. The output voltage of the converter must be maintained at the required DC-link voltage level.

6.2.3 Battery SOC, Current, and Voltage

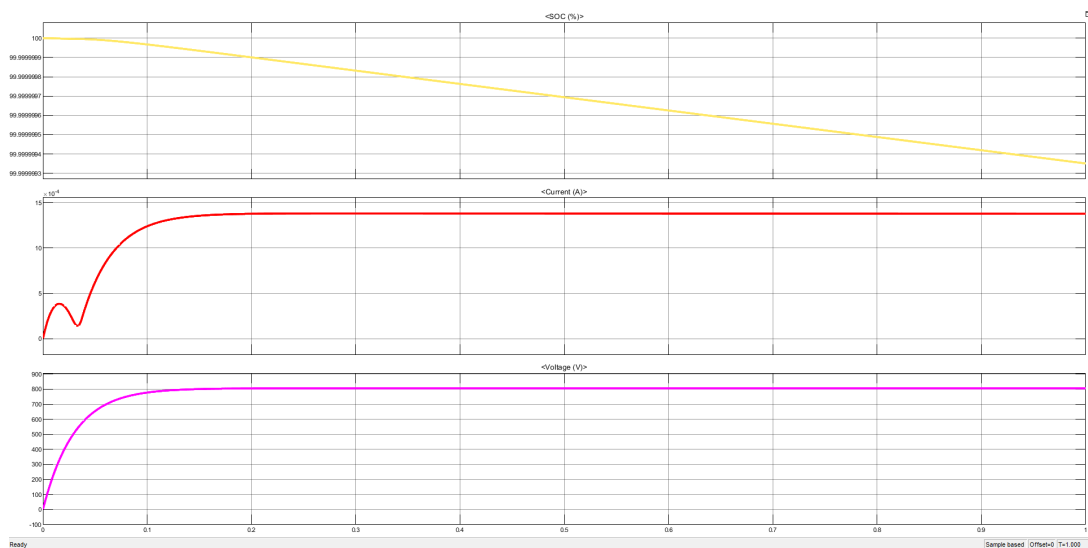


Figure 6.6: State 2: Battery SOC, current, and voltage during discharging

During discharging, the battery SOC decreases gradually. The battery current reverses direction compared to the charging condition. Battery voltage may also reduce depending on load demand and battery internal characteristics.

6.3 State 3: Battery Standstill Condition

In the standstill condition, the battery is neither charging nor discharging. The battery remains idle and no significant power exchange takes place.

6.3.1 Grid-Side Outputs

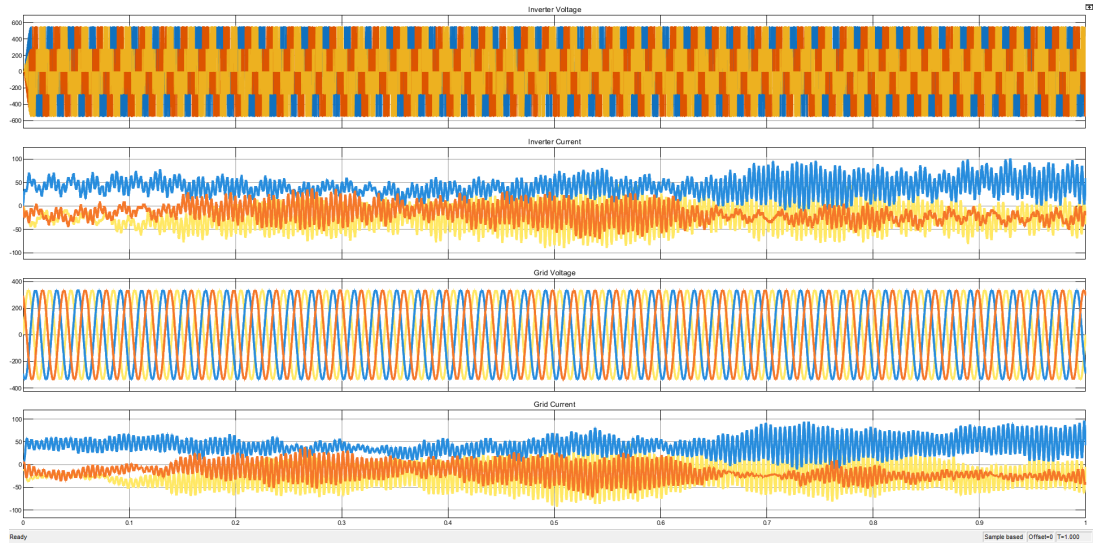


Figure 6.7: State 3: Grid-side voltage and current outputs during battery standstill condition

During standstill mode, the grid-side current is expected to reduce significantly because no active power exchange takes place between the battery and the grid.

6.3.2 Boost Converter Outputs

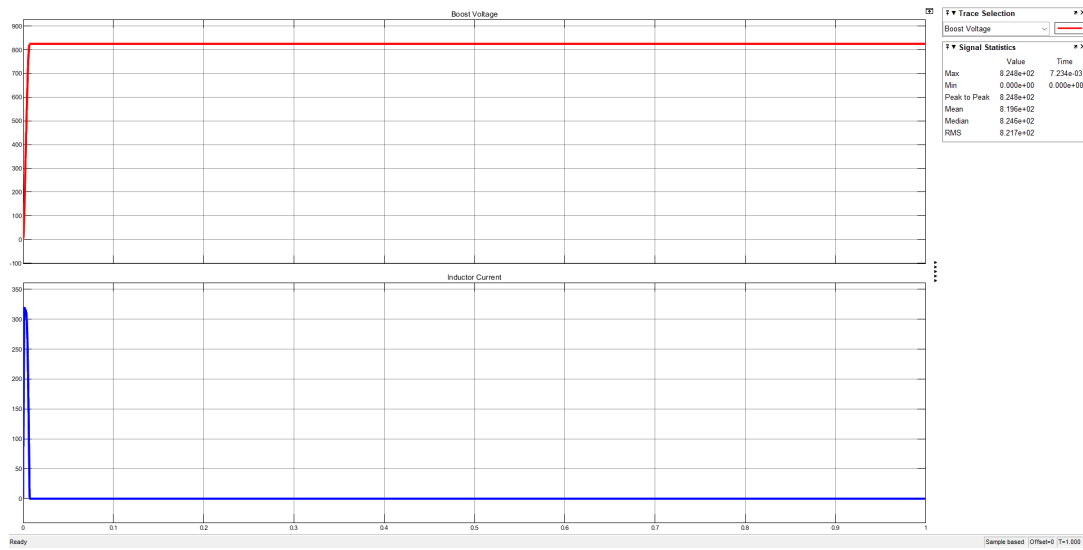


Figure 6.8: State 3: Boost converter voltage and current outputs during battery standstill condition

The converter remains inactive or operates at minimum power transfer. The DC-link voltage may remain constant depending on the control strategy and system losses.

6.3.3 Battery SOC, Current, and Voltage

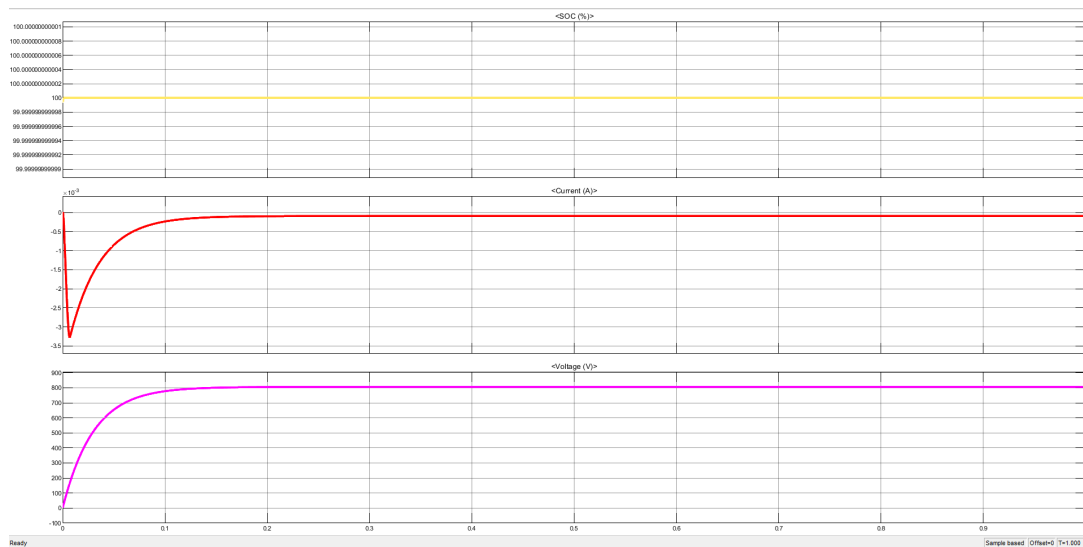


Figure 6.9: State 3: Battery SOC, current, and voltage during standstill condition

In standstill mode, the battery current is approximately zero. The battery SOC remains almost constant because no charging or discharging occurs. Battery voltage also remains nearly steady.

Chapter 7

Observations

The following observations were made from the simulation:

- In State 1, the battery charges from the grid and the SOC increases.
- In State 2, the battery discharges and supplies power to the grid or load.
- In State 3, the battery remains in standstill condition with nearly zero current.
- The boost converter regulates the power flow between the battery and DC-link.
- The battery voltage and current vary depending on the operating state.
- The grid current and inverter current were not synchronized properly.
- Lack of synchronization can result in poor grid power quality and unstable power transfer.
- dq-axis control is required for proper grid-connected inverter operation.

Chapter 8

Need for dq Control

For a grid-connected inverter, the inverter current must be synchronized with the grid voltage. In the present simulation, the grid current and inverter current were not synchronized, which indicates that a more advanced control strategy is required.

The dq-axis control method is commonly used for three-phase grid-connected systems. It uses a Phase Locked Loop to detect the grid voltage angle. The three-phase currents are transformed into dq-axis components using Park transformation.

The transformation from three-phase quantities to dq quantities is given by:

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos \left(\theta - \frac{2\pi}{3} \right) & \cos \left(\theta + \frac{2\pi}{3} \right) \\ -\sin \theta & -\sin \left(\theta - \frac{2\pi}{3} \right) & -\sin \left(\theta + \frac{2\pi}{3} \right) \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

In the dq reference frame:

- The d-axis current controls active power.
- The q-axis current controls reactive power.

For unity power factor operation:

$$i_q^* = 0$$

The active power flow is controlled by adjusting:

$$i_d^*$$

Therefore, dq control improves synchronization, current regulation, power factor, and grid stability.

8.1 dq Control Implementation for Grid Synchronization

For proper grid integration, the inverter output current must be synchronized with the grid voltage. In the open-loop simulation, it was observed that the inverter phase currents and the grid-side currents were not properly synchronized. The phase currents showed a mismatch in phase with respect to the grid voltage, which indicates that open-loop inverter operation is not sufficient for grid-connected operation.

To overcome this issue, a closed-loop synchronous reference frame control strategy, commonly known as dq control, is required. In dq control, the three-phase quantities are transformed into a rotating reference frame using Park transformation. This allows the AC quantities to be converted into DC quantities, making them easier to regulate using PI controllers.

The main objective of dq control is to control the active and reactive power exchange between the inverter and the grid. The direct-axis current component, i_d , is generally used to control active power, while the quadrature-axis current component, i_q , is used to control reactive power.

$$P = \frac{3}{2}V_d i_d$$

$$Q = -\frac{3}{2}V_d i_q$$

For unity power factor operation, the reactive power reference is set to zero. Therefore, the reference quadrature-axis current is selected as:

$$i_q^* = 0$$

The active current reference i_d^* is generated based on the required active power transfer from the DC-link or battery energy storage system to the grid.

8.1.1 Closed-Loop dq Control Structure

The closed-loop control system consists of a phase-locked loop, abc-to-dq transformation, current control loops, dq-to-abc transformation, and PWM pulse generation for the inverter. The PLL is used to estimate the grid angle θ , which is required for reference frame transformation.

The basic closed-loop control structure is shown in Figure 8.1.

The direct-axis controller controls active power transfer, while the quadrature-axis controller controls reactive power transfer. For grid-connected operation at unity power factor, i_q^* is set to zero.

8.1.3 Grid-Side Voltage and Current Observation

The grid-side voltage and current waveforms were observed to analyze the synchronization between the inverter and the grid. Ideally, for unity power factor operation, the grid current should be sinusoidal and in phase with the corresponding grid voltage.

The grid side waveforms is shown in Figure 8.2.

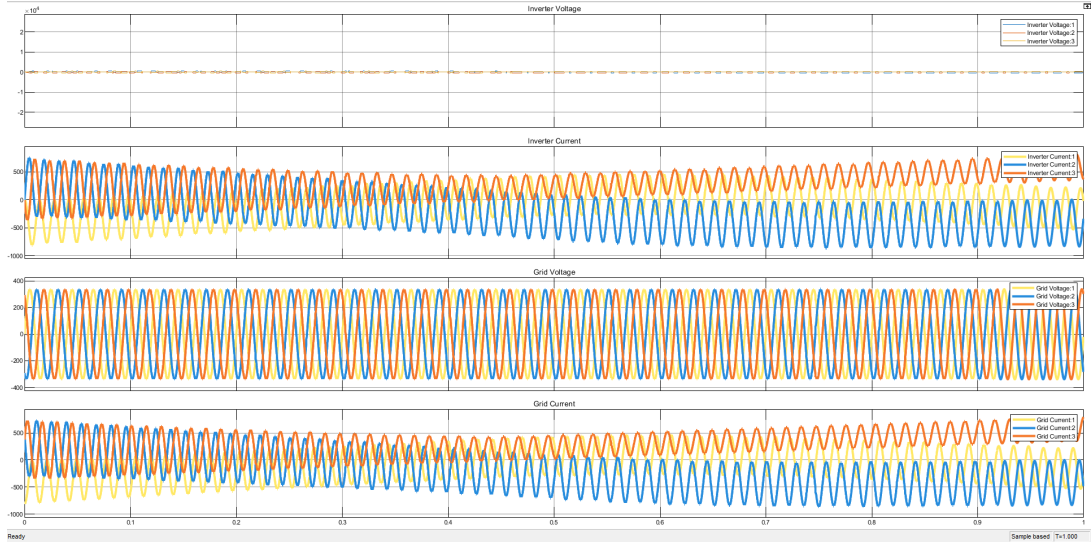


Figure 8.2: Grid-side three-phase voltage waveform

From the simulation results, it was observed that the inverter-side current and grid-side current were not properly synchronized. The phase currents were not in synchronism with the grid voltage, indicating that the inverter was not correctly locked to the grid. This mismatch can lead to poor power transfer, circulating currents, reactive power injection, and unstable grid-connected operation.

Therefore, implementation of PLL-based dq control is necessary for proper synchronization. The PLL ensures that the inverter reference frame follows the grid voltage angle, while the dq current controller regulates the inverter current according to the desired active and reactive power references.

8.1.4 Reason for Lack of Synchronization

The lack of synchronization between inverter current and grid current may occur due to the following reasons:

- Absence of a properly tuned PLL for grid angle detection.

- Incorrect abc-to-dq or dq-to-abc transformation angle.
- Improper PI controller gains in the dq current control loop.
- Lack of current feedback control in the inverter.
- Phase sequence mismatch between inverter output and grid voltage.
- Incorrect PWM reference generation.
- Insufficient filtering at the inverter output.

Hence, the inverter cannot be operated in open-loop mode for reliable grid integration. A closed-loop dq control strategy is required to ensure that the inverter currents are synchronized with the grid voltages.

8.1.5 Expected Result After dq Control

After proper implementation of dq control, the grid voltage and current should become synchronized. For unity power factor operation, the current waveform should be in phase with the corresponding phase voltage.

$$V_a \parallel I_a$$

$$V_b \parallel I_b$$

$$V_c \parallel I_c$$

The expected result is that the inverter injects balanced sinusoidal currents into the grid with controlled active power and minimum reactive power. This improves grid stability and enables proper power transfer from the battery energy storage system to the grid.

Chapter 9

Applications

Battery Energy Storage Systems with grid integration are used in:

- Renewable energy integration
- Solar and wind energy storage
- Peak load shaving
- Grid frequency support
- Backup power supply
- Microgrid applications
- Electric vehicle charging stations
- Uninterruptible power supply systems

Chapter 10

Advantages and Limitations

10.1 Advantages

- Allows bidirectional power flow between battery and grid.
- Improves energy management.
- Supports renewable energy integration.
- Provides backup power during grid disturbances.
- Can improve grid stability when properly controlled.

10.2 Limitations

- Requires accurate control for grid synchronization.
- Battery performance depends on SOC, temperature, and aging.
- Power electronic converters introduce switching losses.
- Improper synchronization can cause poor power quality.
- dq control and PLL design increase control complexity.

Chapter 11

Conclusion

A Battery Energy Storage System with grid integration was designed and simulated. The system was analyzed under three operating states: charging, discharging, and standstill condition.

During the charging state, power was transferred from the grid to the battery and the battery SOC increased. During the discharging state, power was supplied from the battery to the grid and the SOC decreased. During the standstill condition, the battery current was nearly zero and the SOC remained almost constant.

The results showed that the converter and battery behavior changed according to the operating state. However, it was observed that the grid current and inverter current were not synchronized properly. This indicates that dq-axis control is required for accurate grid synchronization and stable power exchange.

Therefore, the next stage of this project should focus on implementing dq-axis control with a Phase Locked Loop for the grid-connected inverter.